

Exploring Tactile Perception: Development and Evaluation of the PinArray, a Novel Haptic Device

Iliyas Tursynbek , John de Grosbois , Brendan Hart, and Mounia Ziat 

Abstract—Despite advances in vibrotactile displays, most existing systems are limited in their ability to deliver calibrated, frequency-differentiated stimulation across multiple touch modes. This constrains our understanding of how supra-threshold frequency modulation influences tactile perception, particularly in dynamic, shape-based interactions. To address this gap, we introduce the PinArray—a novel hybrid haptic device featuring a 4×3 array of independently actuated pins capable of delivering vibrations from 0 to 300 Hz. The PinArray uniquely supports static, passive, and active touch conditions, enabling nuanced exploration of tactile shape encoding. We evaluated the device in a user study examining the perception of edge-like shapes generated via frequency pairings. Results show that specific combinations, especially those involving static and dynamic frequency pairs, significantly enhance shape recognition. These findings highlight the device’s potential for advancing both perceptual research and the development of expressive tactile interfaces.

Index Terms—Hybrid haptic device, vibrotactile display, perception and psychophysics.

I. INTRODUCTION

THE sense of touch, also known as somatosensation, is fundamental to human interaction, enabling us to perceive textures, shapes, and other physical attributes through skin receptors distributed throughout the body. This rich sensory modality underpins essential aspects of daily life, from grasping objects to conveying emotions through physical contact [1]. The hands—particularly the fingertips—are uniquely equipped with a high density of cutaneous mechanoreceptors [2], making them critical for fine tactile perception and complex interactions with the physical world. Four main mechanoreceptor types are responsible for sensing a wide range of frequencies: Merkel

disks (SAI) and Ruffini cylinders (SAII), which respond to slow, sustained pressure, and Meissner corpuscles (RAI) and Pacinian corpuscles (RAII), which respond to rapid vibrations [3]. These receptors are typically associated with specific frequency ranges [4], supporting the perception of various tactile sensations such as texture, vibration, and pressure [5]. However frequency-specific studies focus on threshold levels of perception [6], [7], leaving a gap in understanding how frequency modulation impacts perception at supra-threshold levels [8], [9]—critical for designing effective haptic devices that simulate a range of real-world tactile experiences.

Existing vibrotactile displays have contributed significantly to our understanding of touch. Early systems, such as the Optacon [10], and more recent devices like those by Kim et al. [11], enabled independent control of multiple vibrating pins. However, many of these devices are limited by narrow frequency ranges, uniform pin behaviors, or are designed primarily for passive interaction. Few are calibrated to precisely deliver stimuli across the entire 0–300 Hz range or to explore the nuanced effects of frequency differentials on tactile edge perception. Moreover, support for static, passive, and active touch scenarios within the same device remains rare, leaving a gap in studying how these modes influence perception and interaction. In static touch, the fingertip experiences pressure or vibrations from a stationary object without any relative motion between the object and the finger. Passive touch involves external movement against the skin (i.e., a display sliding against a motionless finger) where the individual does not actively move their finger, hand, or arm, while active touch allows users to dynamically explore objects, typically yielding the most detailed tactile information about the object’s properties [12].

To address this, we present the PinArray (see Fig. 1), a hybrid haptic device, which combines both cutaneous and kinesthetic feedback, designed to investigate tactile perception beyond threshold detection. The PinArray consists of a 4×3 array of independently controlled vibrotactile pins capable of delivering precisely calibrated frequencies (0–300 Hz) across a fingertip-sized contact area. The PinArray supports dynamic frequency modulation across pins, enabling systematic simulation of tactile shapes (e.g., ridges, grooves, edges) under static, passive, and active touch conditions. The system includes x/y tracking, and a custom control interface that allows custom waveform generation and synchronized pin actuation. A detailed user study examines how frequency pairings affect shape recognition accuracy—particularly under supra-threshold stimulation—providing new insights into tactile encoding through frequency. By

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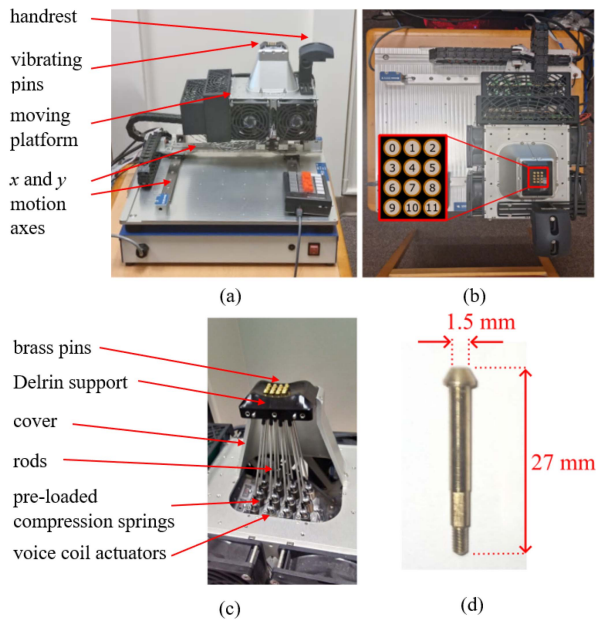


Fig. 1. PinArray device: (a) General structure of the PinArray, (b) Pin numbering used for control, (c) The internal structure of the vibrotactile stimulator (front cover removed for demonstration purposes), (d) Shape and size of a single pin.

integrating both active and passive touch capabilities, the device can simulate tactile sensations under various conditions, expanding its potential applications in sensory augmentation [13], assistive technology [14], and immersive environments such as virtual and augmented reality [15].

The remainder of this paper is organized as follows: Section II reviews existing vibrotactile displays and hybrid haptic systems. Section III outlines the design requirements for the PinArray. Section IV describes the system implementation and calibration. Section V presents an initial user evaluation, and Section VI discusses the results, conclusions, and directions for future work.

II. RELATED WORK

A. Touch Mechanisms in Haptics

In haptic exploration, tactile and kinesthetic receptors work in tandem to convey detailed sensory information [16]. Cutaneous mechanoreceptors, located in the skin, respond to tactile stimuli such as pressure and vibration, while proprioceptors, located in muscles and joints, relay kinesthetic feedback essential for force and motion perception. Together, these receptors enable fine touch discrimination and dynamic object interactions [17]. Research on haptic devices has leveraged these mechanisms to simulate tactile and force feedback, with devices offering either cutaneous, kinesthetic, or hybrid feedback to improve sensory fidelity and user interaction [18], [19], [20], [21], [22].

B. Tactile Displays for the Fingertip

While haptic interfaces often focus on force or kinesthetic feedback, tactile displays emphasize stimulation of the cutaneous sense [23], primarily through arrays of vibrating pins

or tactors that convey texture, shape, and roughness information [24]. These displays operate by activating mechanoreceptors in the skin, primarily through two key design parameters: spatial resolution (pin arrangement) and temporal resolution (vibration frequency). For instance, devices operating within the 100 to 300 Hz frequency range predominantly stimulate rapidly adapting (RA) mechanoreceptors, while lower-frequency devices can engage both slowly adapting (SA) and RA receptors, allowing varied tactile experiences [25].

1) *Resolution Parameters and Perceptual Limits:* Psychophysical studies have shown that humans can reliably distinguish between points separated by as little as 0.9 mm [26], though minimal perceptual gains occur with pin spacing between 1 mm and 2 mm [27]. Standards like Braille, for example, use a spacing of 2.5 mm to balance mechanical design with human perceptual limits [28], [29]. A study by Garcia-Hernandez et al. further explored optimal pin spacing, finding that 1.8 mm within a 1 cm² area provided high accuracy for shape recognition and ridge angle discrimination [30], [31]. This research suggests that pin spacing should consider both mechanical constraints and sensory accuracy to optimize performance.

Alternative configurations, such as diagonal grids with 3.6 mm spacing, have also been effective in conveying complex surface structures [32]. Another approach used hexagonal pin arrangements with skin indentation frequencies between 5 and 80 Hz, demonstrating that spatial layouts must be designed in conjunction with appropriate temporal frequencies to effectively encode tactile information [33]. Lateral movement-based displays, which simulate sliding sensations across the skin, are another avenue for tactile simulation beyond vertical indentation alone [34], [35], [36], [37]. Some designs feature movable, non-vibrating pins on a vibrating surface, adding a layer of sensory complexity [38]. These studies and alternative configurations demonstrate that both pin layout and motion directionality, alongside frequency modulation, play crucial roles in tactile feedback, laying a foundation for the design of higher-fidelity tactile displays.

2) *Early Tactile Displays: Limited Resolution and Bandwidth:* One of the earliest tactile displays was the Optacon, designed for visually impaired users, which converted visual information into tactile sensations. It featured a 24 × 6 matrix of 144 pins actuated by piezoelectric bimorphs vibrating at a fixed frequency of 250 Hz [10], demonstrating the potential of high-frequency stimulation for real-time letter and shape recognition. This pioneering system set the stage for subsequent devices that built on this principle. For instance, a tactile search display with an 8 × 8 pin matrix spaced 5 mm apart was developed for visually impaired users [39], as well as a 6 × 6 pin graphical display with 3 mm spacing for basic shape transmission [40]. Although limited to lower frequencies, these systems established the feasibility of shape recognition through tactile displays.

To enrich tactile experiences, later systems expanded the range of vibration frequencies and refined spatial arrangements. For example, a 1 × 10 array actuated by shape memory alloys enabled low-frequency vibrations from 1 to 40 Hz, facilitating texture simulations through coarse temporal cues [41]. More advanced designs, such as a 4 × 4 piezoelectric ultrasonic

actuator array, achieved a spatial resolution of 1.5 mm with a temporal resolution of 20 Hz, although constrained to low-frequency feedback [42]. Other systems prioritized cost-effectiveness, like a servomotor-actuated 6×6 matrix limited to 7.5–25 Hz due to motor performance [43]. Despite these innovations, many early displays primarily focused on low frequencies (under 100 Hz), which limited their applications in studies requiring fine textures or more complex tactile feedback.

3) *High-Fidelity Displays. Enhanced Resolution and Bandwidth:* Recent advancements in tactile display technology have pushed the boundaries of spatial resolution and bandwidth, enabling more refined and realistic tactile simulations. The HAPTEX system integrates a 6×4 matrix of piezoelectric bimorph-based tactors with a 2 mm spatial resolution and supports dual-frequency operation at 40 Hz and 320 Hz to engage both SA and RA mechanoreceptors [44], [45]. By incorporating force feedback and fingertip tracking, the HAPTEX device enhances the realism of virtual environments and supports intricate tactile tasks.

A modular pin-array device proposed by Kim et al. [11], provides two tactile display configurations (3×3 and 3×4 matrices) with 3 mm pin separation, each capable of producing independent pin vibrations from 0 to 150 Hz—an adaptive range that allows encoding of both fine textures and low-frequency event [46]. Similarly, a 10×10 array with 100 moving pins spaced at 0.6 mm offered a wide operational range of 20–400 Hz, although user testing was limited to 40 and 320 Hz due to perceptual sensitivity and device stability [47].

Among the highest resolution designs is a 20×20 array with 400 pins densely packed within a 1 cm^2 area, capable of frequencies between 0 and 250 Hz [48]. This design, while offering exceptional resolution, is limited in practical application due to its size and complexity. More compact systems—such as a 5×6 array of piezoelectric bimorphs—demonstrated bandwidths up to 400 Hz while balancing resolution with practical usability in perception studies [49], [50]. Across these high-fidelity designs, increased temporal resolution not only expands the repertoire of tactile sensations but also allows for more lifelike simulation of surface interactions and event dynamics.

In summary, the evolution of tactile displays, from early pin-based arrays to high-density, frequency-flexible devices, demonstrates a clear trend toward enhancing both spatial resolution and temporal bandwidth. These advancements lay the groundwork for hybrid haptic devices like the PinArray, which not only offer independent frequency control per pin but also integrate proprioceptive feedback. This combination of cutaneous and kinesthetic feedback opens new possibilities for tactile simulation, supporting richer and more immersive applications in haptic research.

C. Hybrid Haptic Devices: Cutaneous and Proprioceptive Feedback

Hybrid haptic devices integrate vibrotactile and force feedback to enrich the user's tactile experience, providing both cutaneous and proprioceptive information. These systems combine the spatial and temporal resolution of tactile displays—achieved

through pin arrays and modulated vibrations—with kinesthetic movement that enables spatiotemporal navigation across virtual surfaces. This integration enhances perceptual fidelity and supports complex, dynamic interaction. One pioneering example, the TextureExplorer, combines a vibrotactile display with the PHANToM kinesthetic device to support virtual texture rendering. The vibrotactile component consists of 10 pins arranged in a 2×5 matrix with a spatial resolution defined by 3 mm pin spacing and $0.5 \times 0.5 \text{ mm}^2$ pin size. Each pin is independently actuated by piezoelectric elements vibrating at approximately 250 Hz, providing a maximum vertical displacement of 22 microns. The PHANToM delivers 3-DOF force feedback to the user's finger and palm, while the vibrotactile pins target the fingerpad. Studies show that users perceive textures more accurately and efficiently when this hybrid feedback is available, as compared to force feedback alone [51].

Building on this approach, Sarakoglou et al. developed a system that integrates a vibrotactile array. The vibrotactile display includes 16 tactors in a 4×4 matrix with 2 mm center-to-center spacing and a tactor diameter of 1.5 mm, offering high spatial density. These tactors are driven by DC motors pre-loaded with spring tension, achieving vertical displacements of up to 2 mm at 10 Hz and 0.5 mm at 30 Hz. Combined with 7-DOF force feedback from the Omega.7, this hybrid system enabled users to perform teleoperation tasks with smoother trajectories and reduced contact forces, demonstrating the synergistic effect of tactile and kinesthetic feedback [52].

Advancing the use of lateral skin deformation, the Latero device (also known as STReSS²) delivers virtual tactile graphics through an 8×8 matrix of laterally moving contactors. Each contactor, driven by a piezoelectric actuator operating at 50 Hz, contributes to a dynamic tactile image that can be refreshed at 690 Hz. Spatial resolution is determined by the density of the matrix, while proprioceptive input is provided via a mouse-type interface tracked by a 2-DOF pantograph. This combination enables users to explore tactile virtual textures with nuanced cutaneous feedback and intuitive movement control. Notably, user performance remained consistent whether tactile detail was adjusted manually or automatically based on exploration speed, suggesting effective integration of the tactile cues [53].

A related system was proposed by Kyung et al., who developed a mouse-type device incorporating eight piezoelectric bimorph actuators, each controlling a 6×1 pin array for a total of 48 pins. Each pin has a diameter between 0.5 and 0.7 mm, and the spacing can be adjusted between 1 and 2 mm, enabling the simulation of textured surfaces up to 8 mm wide. The lateral translation mechanism, powered by a DC motor and belt-pulley system, supports continuous texture display across a broader surface. When coupled with the PHANToM device, this system was adapted for a medical palpation simulator, providing realistic area-based haptic rendering that improved tactile training for medical professionals [54], [55].

In summary, hybrid haptic devices employ diverse mechanical and actuation strategies to deliver both cutaneous and proprioceptive feedback. By combining spatially distributed tactors with temporally controlled actuation and kinesthetic interfaces that support active exploration, these systems enable rich

TABLE I
SPATIO-TEMPORAL DESIGN PARAMETERS OF VIBROTACTILE DISPLAYS

Author (year)	Pin matrix	Pin spacing (mm)	Contact area (cm ²)	Frequency (Hz)	Actuation	Touch type	Workspace (cm)
Bliss et al. (1970) [10]	24 × 6	Not specified	10.0 × 5.0	12.5-250	Piezoelectric bimorphs	Static, Active	Not specified
Frissen-Gibson et al. (1987) [39]	8 × 8	5.0	3.5 × 3.5	Discrete static height levels	DC solenoids	Static	Not available
Wellman et al. (1988) [41]	10 × 1	2.0	Not specified	1-40	Shape memory alloys	Static, Active	Circular with diameter = 15
Taylor et al. (1997) [32]	Diagonal, 64 tactors	3.6	2.0 × 4.0	62.5	Shape memory alloys	Static	Not available
Caldwell et al. (1999) [35]	4 × 4	1.75	Not specified	0-20 (pressure unit)	Pneumatic actuators	Static	Not available
Kammermeier et al. (2000) [40]	6 × 6	3.0	1.6 × 1.6	100	Electromagnetic rocker levers	Static	Not available
Summers et al. (2002) [47]	10 × 10	0.6	1.0 × 1.0	20-400	Piezoelectric bimorphs	Static	Not available
Ikei et al. (2002) [51]	2 × 5	3.0	Not specified	250	Piezoelectric bimorphs	Static, Active	18 × 18
Wagner et al. (2004) [43]	6 × 6	2.0	1.1 × 1.1	7.5-25	Servomotors	Static, Active	Not specified
Lim et al. (2006) [49]	5 × 6	1.8	0.97 × 0.79	0-400	Piezoelectric bimorphs	Static, Active	Not specified
Wang et al. (2006) [36]	6 × 10	Not specified	1.0	250	Peizoelectric bimorphs	Static	Not available
Magnenat-Thalmann et al. (2007) [44]	6 × 4	2.0	Not specified	40, 320	Piezoelectric bimorphs	Static, Active	40 × 40 × 40
Kim et al. (2007) [55]	6 × 8	1.0-1.5	Not specified	Not specified	Piezoelectric bimorphs	Static, Active	8 × 8
Killebrew et al. (2007) [48]	20 × 20	0.5	1.0 × 1.0	0-250	DC motors	Static, Passive, Active	Not specified
Kim et al. (2009) [42]	4 × 4	1.5	Not specified	1-20	Ultrasonic actuators	Static, Active	Not specified
Garcia-Hernandez et al. (2010, 2014) [30], [31]	5 × 5 - 12 × 12	1.1-3.0	0.4-1.6	Non-vibrating	Indentation across gratings	Static, Passive, Active	12 × 24
Sarakoglou et al. (2012) [52]	4 × 4	2.0	Not specified	7-19	DC motors	Static, Active	10 × 10
Levesque et al. (2012) [53]	8 × 8	Not specified	1.0	50	Piezoelectric actuators	Static, Active	28 × 22
Philpott et al. (2015) [45]	6 × 4	2.0	1.0	40, 320	Piezoelectric bimorphs	Static, Active	10 × 10
Kim et al. (2015) [11]	3 × 4, 3 × 3	3.0	Not specified	0-150	Solenoid type actuators	Static	Not available
Kajimoto et al. (2019) [37]	4 × 8	1.27-1.5	Not specified	10-320	Nichrome wires	Static, Active	Not specified
Grigori et al. (2022) [33]	Hexagonal, 14 tactors	2.5	Not specified	5-80	Voice coil actuators	Static	Not available
Our device—PinArray	4 × 3	4.7	2.0 × 1.5	0-300	Voice coil actuators	Static, Passive, Active	20 × 20

spatiotemporal interaction. This integration enhances perceptual accuracy and task performance across a range of applications. Table I summarizes the key spatio-temporal characteristics of the devices discussed, highlighting their contributions to active touch scenarios.

III. DESIGN REQUIREMENTS

Developing a hybrid haptic device that engages both cutaneous and proprioceptive senses requires a detailed understanding of human tactile perception, as well as the spatial and temporal limits of mechanoreceptors [30], [31], [56], [57]. The

following design requirements guided the development of the PinArray:

- *Pin Array Configuration:* A 4 × 3 matrix that supports spatially distributed tactile feedback, which allows to generate a decent number of tactile stimuli while maintaining cost-effectiveness compared to larger matrix displays. This layout is close in resolution to some of the previous tactile displays [11], [42], [52].
- *Pin Diameter and Spatial Resolution:* Pins have a contact surface diameter of 1.5 mm, with a center-to-center spacing of 4.7 mm to ensure full coverage of the fingerpad (2.0 × 1.5cm). This pin diameter matches the size of

standard Braille pin [28], [29], [58] as well as the tactile display described in [52].

- *Frequency Range*: A range from 0 to 300 Hz to encompass low and high-frequency responses for multiple mechanoreceptors. This allows studying the effects of frequency modulation on touch perception at supra-threshold level.
- *Independent Pin Actuation*: Each pin vibrates independently, allowing for complex frequency combinations across the array and the stimulation with a broader range of receptors.
- *Workspace*: A square area of approximately 20×20 cm to comfortably accommodate the arm movement without causing discomfort, and to suit the testing of both passive and active touch conditions.
- *Control and Data Recording*: A control system with precise x/y displacement positioning and tracking, frequency control, and capability for custom vibration profiles (e.g., sine, square waveforms) to tailor experimental setups. This allows testing several stimuli and experiments and provides relevant data for subsequent analysis.

IV. SYSTEM DESIGN AND CALIBRATION

The PinArray (Fig. 1) consists of 12 independently controlled pins arranged in a 4×3 matrix, each operating within a frequency range of 0 to 300 Hz. This configuration distinguishes the PinArray from conventional vibrotactile devices that typically maintain uniform pin frequencies and are often configured for static touch. Designed as a hybrid vibrotactile stimulator, the PinArray delivers complex tactile stimuli through its arrangement of vibrating pins positioned above two perpendicular linear motion axes. This setup enables controlled movement along the x and y axes, supporting static, passive, and active tactile exploration [59].

A. Hardware Description

1) *Tactile Display*: The tactile display, illustrated in Fig. 1(b), features an array of 12 brass pins shaped as conical frustums with a 4.7 mm center-to-center spacing in their initial position. Each pin has a 1.5 mm diameter circular contact surface (Fig. 1(d)), designed to optimize fingertip contact area. Pins are housed within a curved Delrin plastic cover (Fig. 1(c)), which maintains spacing and alignment while allowing some variation in spacing as pins move. This arrangement yields an initial stimulation surface of approximately 2.0×1.5 cm. For reference, each pin is numbered from 0 to 11.

To maintain fingertip alignment during testing, a 3D-printed hand rest positions the user's finger along the array's longer axis, supporting consistent and gentle contact with the display. Adjustable in four heights (105, 110, 115, and 120 mm), the hand rest allows lateral positioning to suit individual needs.

Each pin vibrates independently, powered by H2W Technologies' NCM02-05-005-4JBL voice coil actuators linked to the pins through rods and compression springs (Fig. 1(c)). The actuators are controlled by individual driver boards with 655 U operational amplifiers, enabling precise frequency control. Regular maintenance with lubricating oil ensures smooth operation.

An ATI Mini40 6-axis force/torque sensor embedded in the tactile display records the force exerted by the participant on the top surface.

2) *Linear Motion System*: The PinArray's robust linear motion system supports weights up to 12 kg and enables movement in four cardinal directions using SSELBWN linear guides. Operating within a 200×200 mm planar workspace, the system provides high positional accuracy with a resolution of 2000 lines per inch, monitored by EM2 optical encoders equipped with EM2-0-2000-I linear strips on each axis. These optical encoders are critical for capturing precise x/y displacement, speed, and acceleration data at a 10 Hz acquisition rate, facilitating detailed analysis of motion parameters.

B. Software Description

The custom-built controller box, essential to the PinArray's functionality, interfaces with an experimental computer via Ethernet. This controller generates PWM signals that are amplified by a motor driver and then transmitted to the voice coil actuators to control pin movements. System commands and configurations are managed through a Python script, developed in Python 2.7, allowing flexible and precise control of each pin's vibration.

Each pin's behavior is set according to parameters specified in a configuration file (.json), providing a flexible framework for adjusting various experimental settings:

- *Wave Type*: Enables selection of sine, square, constant, sawtooth, or triangular waveforms based on specific experimental requirements.
- *Target Frequency*: Specifies the frequency of oscillation for each pin.
- *Setpoint Value*: Determines pin amplitude, with values calibrated for each pin, as detailed in the calibration process (Section IV-C below).

A visual interface developed with the PsychoPy module supports user interaction. Widely used in behavioral research, PsychoPy provides a user-friendly visual guide and records participants' responses, including timing and duration, facilitating comprehensive data analysis.

C. System Performance and Calibration

Due to non-linear nature of voice coil actuators, each one of them requires a unique gain parameter for its controller's amplifier to achieve the desired vibration amplitude for each of testing frequencies. To ensure accurate vibratory output, a comprehensive calibration process is essential. The maximum achievable amplitude decreases at higher frequencies, making it impossible to maintain a uniform amplitude across the full frequency range. Moreover, the amplitudes might slightly vary for the same vibratory frequency across different pins. To address this, a linear relationship between frequency and amplitude is enforced such that their product remains constant at 1.5 for all pins. This constant value ensures consistency during calibration and meaningful amplitudes even at higher frequencies—for example, 0.005 mm at 300 Hz, as shown in Table II.

Pin vibrations are driven using a lookup table approach, where setpoint values, i.e., unit-less amplifier gains, are recorded

TABLE II
FREQUENCIES TESTED (EXCEPT 0 Hz) AND THE CORRESPONDING AMPLITUDES

Frequency (Hz)	Amplitude (mm)	Frequency $\times$ Amplitude
10	0.1500	1.500
20	0.0750	1.500
25	0.0600	1.500
30	0.0500	1.500
40	0.0375	1.500
50	0.0300	1.500
60	0.0250	1.500
70	0.0214	1.498
75	0.0200	1.500
80	0.0188	1.500
90	0.0167	1.503
100	0.0150	1.500
110	0.0136	1.496
120	0.0125	1.500
125	0.0120	1.500
130	0.0115	1.495
140	0.0107	1.498
150	0.0100	1.500
160	0.0094	1.500
170	0.0088	1.496
175	0.0086	1.505
180	0.0083	1.494
190	0.0079	1.501
200	0.0075	1.500
225	0.0067	1.508
250	0.0060	1.500
300	0.0050	1.500

after being determined from the calibration process for each frequency. A Keyence IL-30 laser camera, mounted perpendicularly to each pin, as shown in Fig. 2), captures pin displacement data at a high sampling rate of 10 kHz using an NI-DAQ card (NI 9215). A custom Python script drives each pin with a sine wave at the target frequency and some initial setpoint value, repeating the signal four times for two seconds each, as illustrated in Fig. 3 for Pin 0 at 63 Hz with setpoint value of 1600. The script then calculates the average peak-to-peak amplitude during the middle one-second window of each repetition. Later this amplitude value is used alongside the staircase algorithm [60] to adjust the setpoint values based on whether the measured amplitude is higher or lower than the desired value.

In the staircase algorithm, the initial step size is set to 100. After four reversals, i.e., direction changes in the adjustment, the step size is reduced to 50. After another four reversals, it is halved again to 25. Following four more reversals at this smallest step size, the staircase is considered complete. The average of the last four setpoint values is recorded as the final setpoint for that specific pin and frequency combination. These calibrated setpoints are stored in a lookup table, enabling precise control over each pin's vibration profile. This iterative process ensures high accuracy within the system's operational limits.



Fig. 2. Calibration with the laser camera. Knobs (circled with red) are used to adjust the pointing direction of the camera.

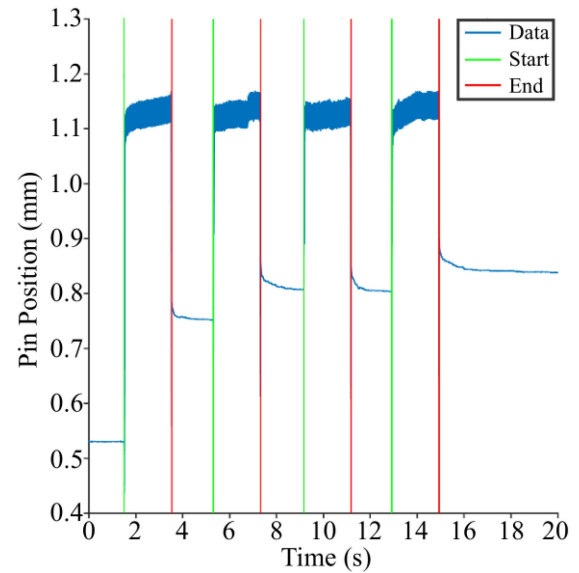


Fig. 3. The data from one trial of the calibration procedure for one of the pins (Pin 0) indicates the start and end of each of the four 2-s vibration durations at a particular frequency (63 Hz) and setpoint value (1600).

V. USER EVALUATION: DETECTING EDGE

To evaluate the PinArray's performance, we investigated the impact of vibration frequency on tactile edge perception by presenting stimuli with varying frequency pairs and shapes. Building on Lim et al.'s research [49], which demonstrated that frequency differentials could simulate tactile edges (Fig. 4), this study broadens the frequency range tested to 0–300 Hz. This wider range allows us to assess the PinArray's effectiveness across a spectrum of tactile scenarios.

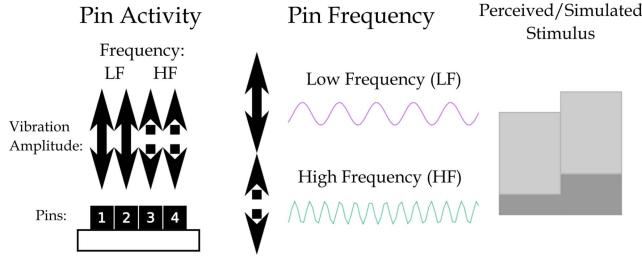


Fig. 4. Simulation of vibrotactile edges through modulating stimulus frequency across an array of pins: region with lower frequency (LF) is perceived as the lowered surface, and region with higher frequency (HF) is perceived as the raised surface.

A. Participants

Forty participants (20 females, 20 males; mean age = 26.13, SD = 7.35) were recruited from the Bentley University community. Participants completed the Edinburgh Handedness Questionnaire [61], with thirty-six identifying as right-handed and four as left-handed. Written informed consent was obtained, and all participants received compensation. None of the participants reported issues with cutaneous or kinesthetic functions. To minimize fatigue, participants were divided into eight subgroups (five participants each), with each group testing specific frequency ranges. This structure ensured that all frequency pairs were covered without overextending session durations, which were kept under one hour. The experimental methods were approved by the Bentley University Institutional Review Board (IRB).

B. Tactile Stimuli and Experimental Groups

The pin display was configured to align with the longitudinal direction of participant's index finger, to accommodate symmetrical stimuli presentation. Each pin could vibrate independently, allowing the simulation of tactile patterns across the fingertip. We tested five shape configurations: ascending and descending step edges (Shape 1 and Shape 2, respectively), groove (Shape 3), ridge (Shape 4), and a flat no-edge surface (Shape 5) as the control condition. As illustrated in Fig. 5, for each shape, frequency pairs (F1, F2) simulated a tactile edge by juxtaposing high (HF) and low (LF) frequencies, with blue dots representing pins operating at LF and red dots pins at HF.

Participants were assigned to one of eight experimental groups, each group evaluating a subset of frequencies with a specific frequency step (10, 25, 50, or 100 Hz). This systematic structure (Fig. 6) enabled the study to capture a range of frequency responses, with Group 1 testing four frequencies (100 Hz step) and the remaining seven groups testing six frequencies (10, 25, and 50 Hz steps). Each configuration included five shapes, totaling 390 trials per participant in each group (162 for Group 1). To minimize auditory distraction, noise-canceling headphones were provided.

C. Procedure

Participants were seated in front of a computer that was interfaced with the PinArray device. The tactile display was positioned to accommodate participant's dominant hand—to

the right for right-handed participants and to the left for left-handed participants. At the start of the experiment, participants were instructed to gently place the index fingertip of their dominant hand on the pins. A hand rest was used to enhance comfort and allow a gentle contact with the PinArray during the experiment. Throughout the study, the experimenter visually monitored participants' hand and finger placement to ensure adherence to this instruction. Directly in front of the participants, a 5-option response keypad was placed, featuring 3D-printed replicas of the shapes to aid in response selection (see Fig. 7).

Participants completed three stages: familiarization, training, and testing.

- **Familiarization Stage:** Participants were simply introduced to the task and the five tactile shapes as shown in Fig. 5, presented in random order to their index fingertip—with their visual representations on a computer screen—without the need for providing their responses. This stage helped them become acquainted with the device and the five stimuli. No information about the frequency range of the stimuli was disclosed.
- **Training Stage:** This stage was similar to the familiarization stage. However, participants were required to identify the five shapes using a response keypad with 3D replicas of each shape. Feedback was provided after each response to help participants calibrate their expectations.
- **Testing Stage:** To ensure auditory distractions did not influence responses, participants wore noise-canceling headphones during this stage. Each participant completed all trials in their assigned frequency step group, with each trial consisting of a 500 ms stimulus followed by a 5-second response window. Missed responses were reintroduced later in the session to ensure data completeness. To reduce fatigue, participants took brief 2-minute breaks throughout the one-hour session.

D. Data Analysis

To analyze the effects of frequency and shape on tactile perception, Linear Mixed Models (LMM) were employed due to their suitability for complex, repeated-measures designs involving both fixed and random effects.

1) **Model Selection and Variables:** LMMs were chosen to handle the intra-participant variability and the influence of repeated trials. The model included two fixed effects: shape (five levels) and frequency pair (specific high and low frequency combinations), as well as a random effect for participants to account for individual differences in sensory perception and response style.

- **Fixed Effects:**
 - **Shape:** Different shapes (e.g., ascending and descending edges, ridge) were expected to yield distinct perceptual responses due to variations in tactile complexity.
 - **Frequency Pair:** Specific combinations of high and low frequencies were predicted to impact edge perception, with particular interest in how frequency differentials would simulate tactile boundaries.

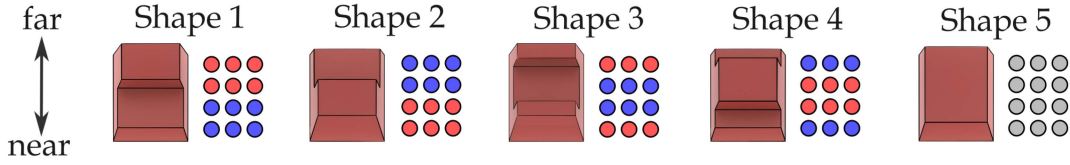


Fig. 5. Five shape configurations: Shape 1 - an ascending step edge (LF to HF), Shape 2 - a descending step edge (HF to LF), Shape 3 - a groove (HF-LF-HF), Shape 4 - a ridge (LF-HF-LF), and Shape 5 - a flat no-edge surface (all pins vibrate at the same frequency). Blue dots represent LF operating pins and red dots represent HF pins.

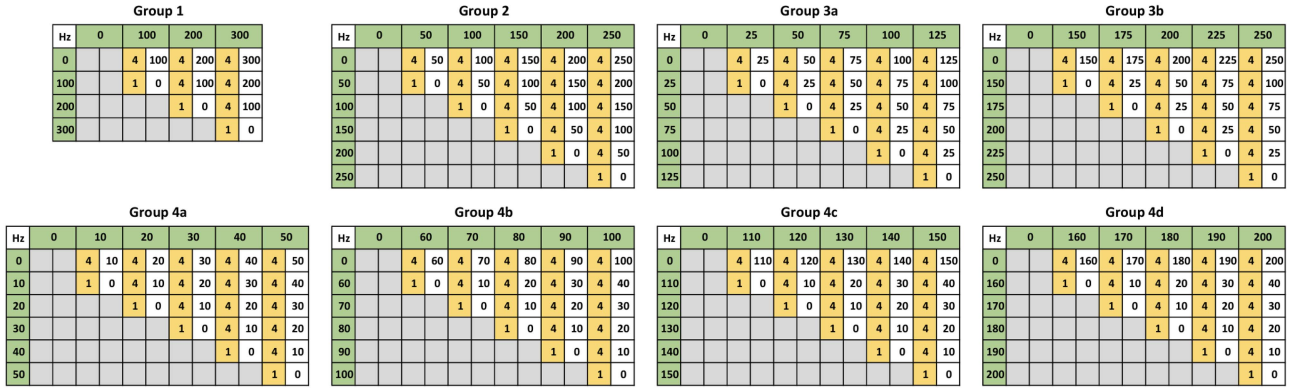


Fig. 6. Experimental Groups: combination of frequencies with 100 Hz step (Group 1), combination of frequencies with 50 Hz step (Group 2), combination of frequencies with 25 Hz step (Groups 3a and 3b), combination of frequencies with 10 Hz step (Groups 4a, 4b, 4c, 4d). Yellow cells represents the number of shapes within each frequency pair. White cells indicates Δf : frequency difference between HF and LF.

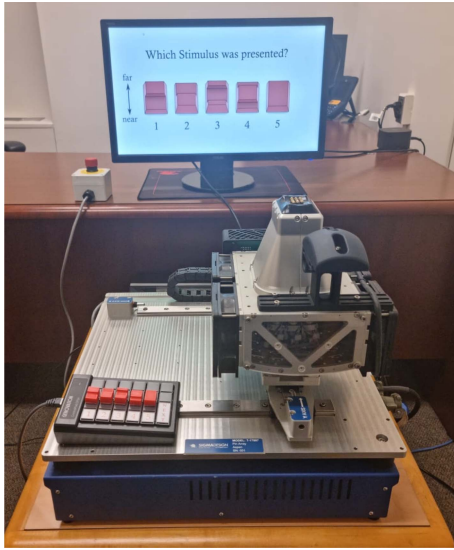


Fig. 7. The experimental setup with the five tested shapes (front view).

- *Random Effects: Participant:* Participant responses were treated as a random effect to control for individual variability and enable generalizable results across the sample.

E. Results

The analysis included data from 2,410 observations across 40 participants. Using the REML criterion (converged at 21221.7), the LMM revealed significant findings on shape and frequency effects:

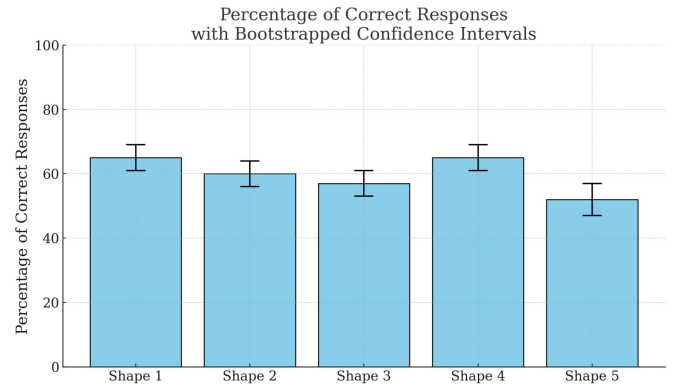


Fig. 8. Percentage of correct responses per shape with bootstrapped (N=1000) confidence intervals.

1) *Shape Distinction:* Shapes 1 and 4 (ascending edge and ridge) achieved the highest identification rates, suggesting that certain tactile configurations are more salient. In contrast, Shape 3 (groove) showed the lowest identification accuracy across all frequency pairs, indicating it may be less effective in simulating edges using the PinArray (Fig. 8). Compared to the reference shape (Shape 1), the model revealed that Shape 2 (descending step edge) was associated with a 4.17 percentage point decrease in correct responses ($p = .002$), while Shape 3 resulted in a 6.16 percentage point decrease ($p < .001$).

2) *Single Frequency Presentation:* The control shape 5 tested with single frequencies revealed that 125 Hz produced a 72.8% correct response rate, significantly outperforming other frequencies in simulating a flat surface ($p < .05$). Frequencies

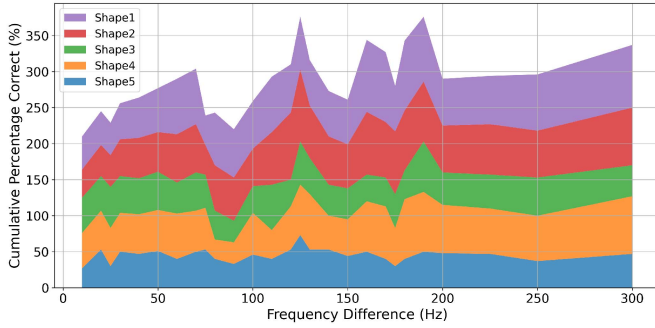


Fig. 9. Stacked area chart showing the percentage of correct responses for five shapes (Shape 1 to Shape 5) across increasing frequency differences (10 Hz to 300 Hz). Each layer represents a distinct shape, and the height of each segment reflects its contribution to the overall performance.

like 200 Hz (49.8% accuracy) also showed moderate efficacy, whereas other frequencies yielded correct responses near the baseline rate of 20%, indicating minimal perception for shapes without distinct frequency differentials.

3) *Frequency Difference*: Participants' ability to identify five different shapes was examined across a range of frequency differences (10 Hz to 300 Hz), with accuracy aggregated across all groups. Accuracy generally increased as the frequency difference grew, with noticeable improvements beginning around 60 Hz and plateauing near 125 Hz for most shapes. As shown in Fig. 9, Shape 1 and Shape 4 showed the most consistent gains, reaching over 80% accuracy at frequency differences of 125 Hz and higher. Shape 2 demonstrated steady improvement, particularly between 100 Hz and 200 Hz, while Shape 5 exhibited moderate accuracy with less sensitivity to changes in frequency. Shape 3 yielded the lowest performance, with modest accuracy across all frequency levels and rarely exceeding 60%. A performance dip was observed between 75 Hz and 90 Hz for several shapes, suggesting difficulty distinguishing frequency pairs within this range.

4) *Frequency Pair. Impact of Zero Frequency*: Frequency pairs involving 0 Hz (i.e., static component) and low HF components (10–20 Hz) achieved correct response rates of 94.7% and 96.6%, respectively, underscoring the perceptual importance of combining static with low-frequency stimuli. As HF increased beyond 50 Hz, perceptual accuracy declined, with moderate recoveries at 180 Hz (73.6%) and 190 Hz (76.9%) before tapering off at 300 Hz (68.2%), as shown in Fig. 10. These results support that static-dynamic frequency combinations effectively simulate tactile edges.

VI. DISCUSSION AND CONCLUSIONS

A. Perceptual Implications

This study investigated how vibrotactile frequency modulation can be used to simulate tactile edges and shapes, with a specific focus on how perceptual performance changes as a function of frequency difference. The PinArray, our novel 4×3 vibrotactile display, enabled the precise control of frequency at the pin level and supported a wide stimulation bandwidth

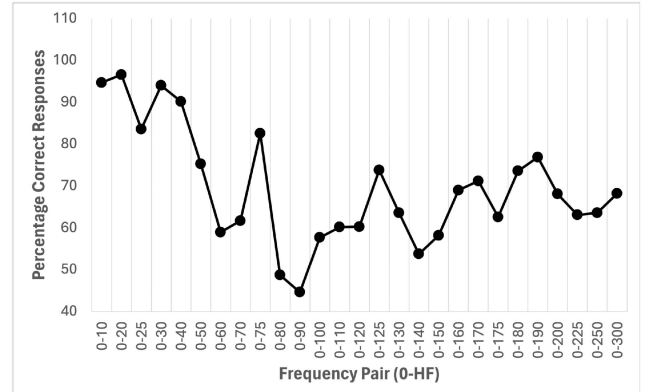


Fig. 10. Percentage of correct responses for each frequency paired with zero (static component).

(0–300 Hz), which allowed us to test edge perception under realistic supra-threshold conditions. Our findings reinforce and extend prior psychophysical work on tactile edge perception [49] and provide important new insights into the perceptual salience of frequency differentials across multiple shape types.

The tactile system's response to vibrotactile stimuli is largely mediated by rapidly adapting mechanoreceptors—RAI (Meissner) and RAI (Pacinian) corpuscles—each with different frequency sensitivities [3], [4]. RAIs typically respond to low-frequency vibrations (3–40 Hz), while RAIs are tuned to higher frequencies (above 60 Hz). Our results show that frequency pairings combining static or low-frequency components (0–20 Hz) with mid- to high-frequency stimuli (e.g., 100–200 Hz) resulted in the highest shape recognition accuracy. This suggests that perceptually salient edges can be simulated by activating mechanoreceptors with distinctly different temporal tuning properties on adjacent pins. The strong performance of combinations involving 0 Hz likely reflects the stark contrast between complete absence of vibration (simulating static contact or pressure) and dynamic activation of RAIs.

Moreover, the observed dip in performance between 75–90 Hz across several shapes may correspond to a transitional range between RAI and RAI sensitivity, where neither receptor type is optimally stimulated. This aligns with neurophysiological literature that identifies a frequency “gap” or impingement in perceptual salience [9], and further supports the idea that maximal perceptual contrast arises when different mechanoreceptor populations are differentially engaged.

Shape-specific differences in recognition accuracy—particularly the high performance of ascending edges (Shape 1) and ridges (Shape 4), and the low performance of grooves (Shape 3)—suggest that some configurations generate more interpretable perceptual patterns than others. These differences likely arise from the directionality and steepness of frequency transitions across the array. For example, edges with a clear unidirectional frequency increase (e.g., Shape 1) may create a stronger perceptual gradient across the fingertip, whereas grooves (HF-LF-HF) may generate competing signals that cancel out spatial differentiation.

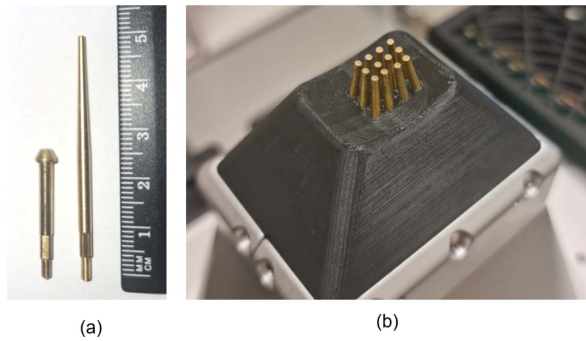


Fig. 11. (a) Comparison between an old and new pin, (b) Vibrotactile stimulator with new pins.

The finding that 125 Hz was optimal in single-frequency trials (Shape 5) for eliciting a flat surface perception further supports the critical role of RAII mechanoreceptor tuning in conveying a “neutral” tactile signal. Frequencies that deviated from this optimal range either lacked sufficient dynamic stimulation (too low) or became perceptually noisy (too high), reducing classification accuracy.

B. Device Contributions and Future Directions

The PinArray distinguishes itself from prior devices not merely by offering independent frequency control—an existing feature in some modular systems [11]—but through its integration of:

- A wide operating range (0–300 Hz), enabling the stimulation of both low- and high-frequency mechanoreceptors.
- A platform designed for static, passive, and active touch (though only static was used in the current study).
- A lookup-based calibration system ensuring consistent amplitude across pins and frequencies.

To further explore the PinArray’s potential, a modified pin array was designed and manufactured with a smaller pin diameter and increased spatial resolution. Specifically, the new pins, each with a 1.5 mm contact surface diameter and 2.5 mm spacing, increase tactile precision (see Fig. 11 for comparison between original and new pins). This improvement supports the tactile differentiation of smaller patterns, as demonstrated in recent tests [62]. While the original pin spacing of 4.7 mm provided adequate results, this new configuration offers enhanced spatial sensitivity, allowing the PinArray to simulate a broader array of tactile shapes. It also enabled us to systematically evaluate how frequency combinations affect perceptual shape recognition at supra-threshold levels. Notably, the integration of force sensing and movement tracking, while not fully utilized in the current study, positions the device for future multimodal interaction studies, particularly in virtual and augmented environments.

C. Limitations and Conclusion

The current study focused exclusively on static touch to isolate frequency-based effects. Future work will incorporate passive and active exploration to assess how proprioceptive cues

interact with frequency-based tactile information. The PinArray’s hardware supports such extensions, and incorporating dynamic finger movement may yield new insights into the integration of cutaneous and kinesthetic feedback. Additionally, extending this work to include more complex surface topographies (e.g., corners) could further test the generalizability of frequency-based encoding strategies. Additionally, this study validates the PinArray as a powerful tool for investigating frequency-specific tactile perception during movement. The modifications in pin diameter and spatial resolution enhance its utility in simulating tactile shapes, making it a valuable resource for future studies in tactile perception, sensory augmentation, and assistive haptic technologies.

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